

Class 08: Breast Cancer Mini Project

Derek Zhang - PID A17819201

Table of contents

Background	1
Data Import	1
Principal Component Analysis Preparation	4
Interpreting PCA results	6
Variance Explained	8
Communicating PCA Results	10
Hierarchical Clustering	11
Visualizing Hierarchical Cluster Results	11
Selecting number of clusters	12
Combining Methods	13
Sensitivity/Specificity	14
Prediction	14

Background

The goal of today's work is to explore a complete analysis using the unsupervised learning techniques covered in class. Analysing a data-set describing the characteristics of cell nuclei from fine needle aspiration of breast mass will reveal critical information on its state of tumor progression.

Data Import

First we will import our data from Wisconsin Cancer data as a dataframe

```
fna.data <- "WisconsinCancer.csv"
wisc.df <- read.csv(fna.data, row.names=1) #runs a csv read into df
```

Next we will visualize this data just to gain a preliminary understanding

```
head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
84358402	0.1809	0.05883	0.7572	0.7813	5.438
843786	0.2087	0.07613	0.3345	0.8902	2.217

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663

84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
84358402	0.4000	0.1625	0.2364	
843786	0.5355	0.1741	0.3985	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			
84358402	0.07678			
843786	0.12440			

To ensure we don't keep the "answer" to our analysis in the database, we must remove the diagnosis column, and we can store it elsewhere.

```
diagnosis <- wisc.df[,1]
wisc.data <- wisc.df[,-1]
```

Q1. There are 569 observations.

```
dim(wisc.data)
```

```
[1] 569 30
```

Q2. 212 observations had the Malignant diagnosis

```
sum(diagnosis == "M") # converts diagnosis into a true or false where true is
```

```
[1] 212
```

```
#malignant and utilizes those values to find how many diagnosis are malignant
```

Q3. 10 variables/features are suffixed with "__mean"

```
length(grep("_mean", colnames(wisc.data), value = T)) # uses a regex pattern
```

```
[1] 10
```

```
# matching method called grep to determine how many col titles have _mean
```

Principal Component Analysis Preparation

First we must check if the data needs to be scaled at all

```
# Check column means and standard deviations
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01

perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

Next, the PCA is executed by running `prcomp()`

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. 0.4427 Proportion of variance is captured by the first principal component.

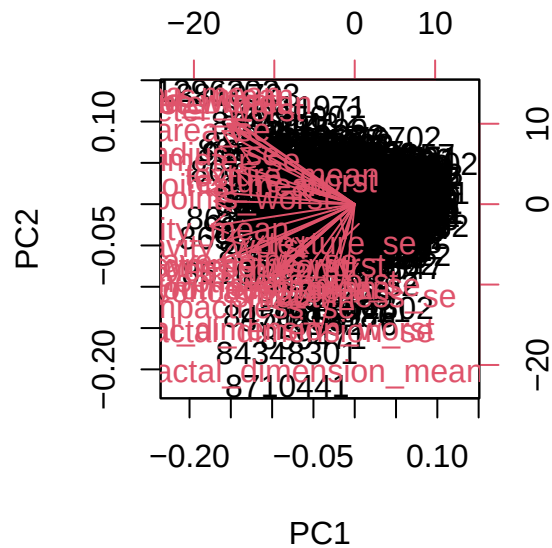
Q5. 3 Principal components, as principal component 3 is when the cumulative proportion sums to over 0.70.

Q6. 7 Principal components, as principal component 7 is when the cumulative proportion sums to over 0.90.

Interpreting PCA results

Q7. There is a lot going on in this graph, seems to have a lot of data that is all overlapping, heavily connecting to one another. Not useful for data.

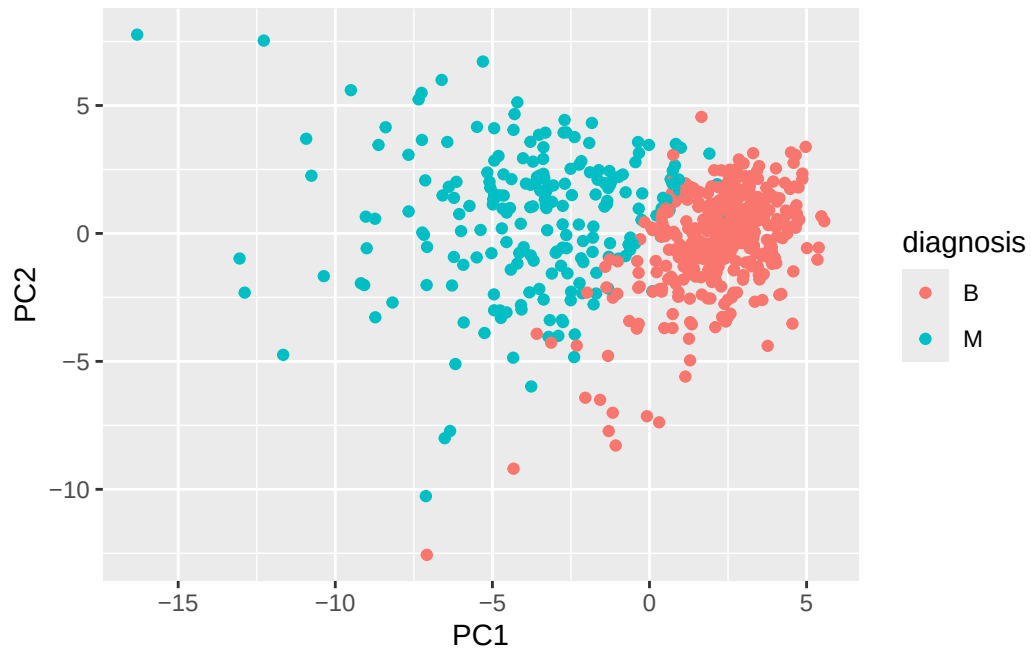
```
biplot(wisc.pr)
```



This is a hard to read and not very useful graph, instead:

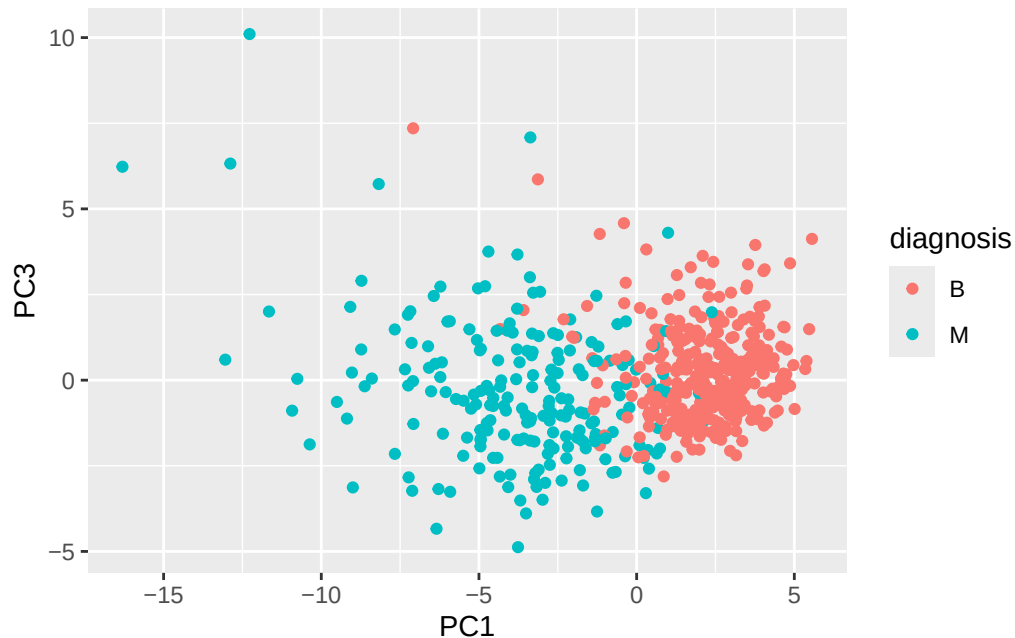
```
# Scatter plot observations by components 1 and 2
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC2, col=diagnosis) + #coloring based on diagnosis result
  geom_point()
```



Q8. A similar graph for PC1 and PC3 shows that benign and malignant observations have distinct separation in component values, which suggests that principal components can be used to diagnose malignancy in samples.

```
# Repeat for components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



Variance Explained

We have to calculate the variance of each PC

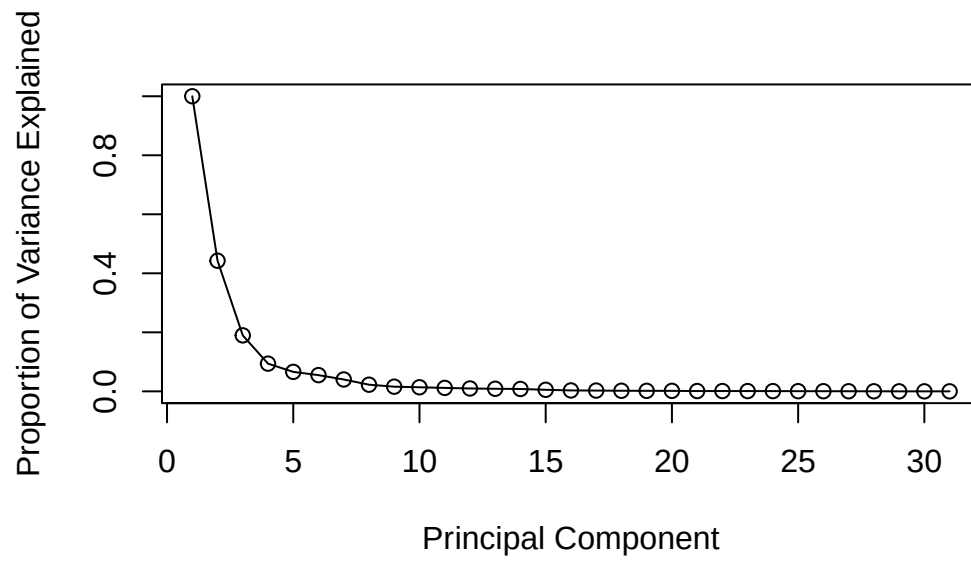
```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

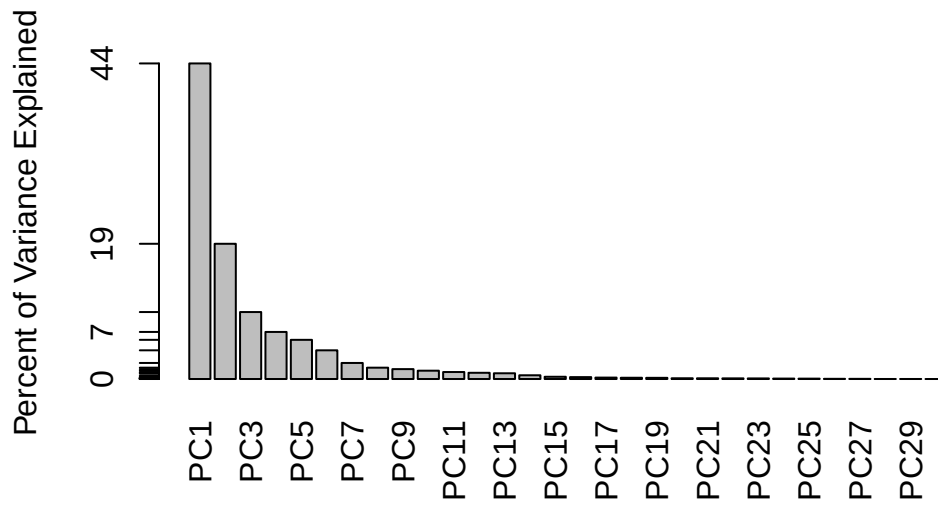
Now we will calculate the variance explained by each PC by dividing total variance explained of all principal components.

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var) #simple division to get ratios of prVar to total var

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA Results

Q9.concave.points_mean for principal component 1 is -0.2609. This feature has the highest absolute value, which means that it is the greatest contributing feature to the first principal component.

```
wisc.pr$rotation[,1] #allows visualization of loading values for PC1
```

radius_mean	texture_mean	perimeter_mean
-0.21890244	-0.10372458	-0.22753729
area_mean	smoothness_mean	compactness_mean
-0.22099499	-0.14258969	-0.23928535
concavity_mean	concave.points_mean	symmetry_mean
-0.25840048	-0.26085376	-0.13816696
fractal_dimension_mean	radius_se	texture_se
-0.06436335	-0.20597878	-0.01742803
perimeter_se	area_se	smoothness_se
-0.21132592	-0.20286964	-0.01453145
compactness_se	concavity_se	concave.points_se
-0.17039345	-0.15358979	-0.18341740
symmetry_se	fractal_dimension_se	radius_worst
-0.04249842	-0.10256832	-0.22799663

texture_worst	perimeter_worst	area_worst
-0.10446933	-0.23663968	-0.22487053
smoothness_worst	compactness_worst	concavity_worst
-0.12795256	-0.21009588	-0.22876753
concave.points_worst	symmetry_worst	fractal_dimension_worst
-0.25088597	-0.12290456	-0.13178394

Hierarchical Clustering

Scaling the data first for clustering

```
data.scaled <- scale(wisc.data)
```

Then calculate euclidean distance between all pairs of observations in the new scaled dataset

```
data.dist <- dist(data.scaled) #creates the distance of data
```

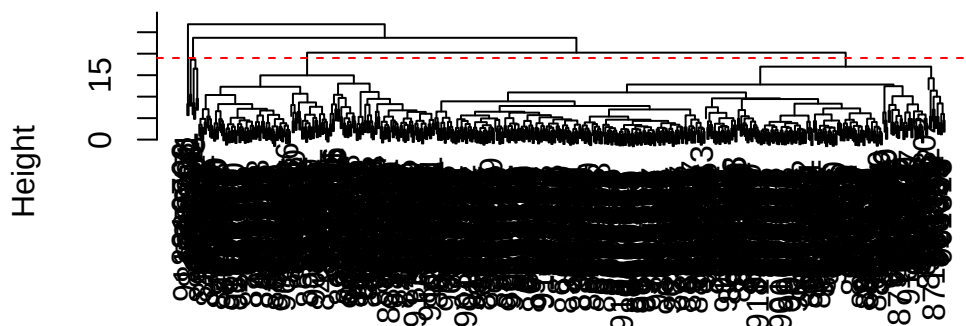
Then create the hierarchical clustering model

```
wisc.hclust <- hclust(data.dist, method="complete") # generates an hclust using
#the method complete
```

Visualizing Hierarchical Cluster Results

```
plot(wisc.hclust) #visualises the hclust
abline(h=19, col="red", lty=2) # adds a line to the previous plot at height 19
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

Q10. The height at which the clustering model has 4 clusters is 19, but that isn't the exact height it starts at, as that would be a non-integer. I do know that at height 19 there are exactly 4 clusters, though.

Selecting number of clusters

Creating a hierarchical cluster with 4 clusters

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4) #cutree cuts the prior tree
# into the desirable number of clusters
table(wisc.hclust.clusters, diagnosis) #visualizes the result of cutting based on
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

```
# diagnoses outcomes, demonstrates cluster correlation to diagnosis
```

Q12. ward.D2 is preferred because it minimizes within-cluster variance, producing more compact and evenly sized clusters compared to the other methods. Single linkage tends to produce chaining effects, while complete and average linkage produce less balanced clusters.

Combining Methods

```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]), method="ward.D2") #dist is important
#to get the distances of the first 7 PCs, using method "ward.D2"
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2) #cut at 2 clusters
grps <- cutree(wisc.pr.hclust, k=2) # creates groups for later
```

Q13. The hclust model separates M and B diagnoses reasonably well. Cluster 1 is predominantly malignant (188 M, 28 B) and cluster 2 is predominantly benign (329 B, 24 M). This method is not perfect as 52 samples were misclassified.

```
table(wisc.pr.hclust.clusters, diagnosis) # outputs a table for comparison
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	28	188
2	329	24

Q14. The non-PCA hclust model performs slightly worse overall. It correctly identifies more benign cases (343 vs 329) but it misses more malignant cases, as it only captured 165 out of 212 M correctly compared to 188 with PCA. Since missing malignant diagnoses is more dangerous in a clinical setting, the PCA-based hclust model performs better in the most important category. As such, because they have approximately the same amount of errors, the PCA-based hclust model is the better performing method.

```
table(wisc.hclust.clusters, diagnosis) # outputs a table for comparison
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Sensitivity/Specificity

Optional question here, will leave blank for now.

Prediction

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url) # reads csv from url
npc <- predict(wisc.pr, newdata=new) #predicts from the previous data, and the new
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029

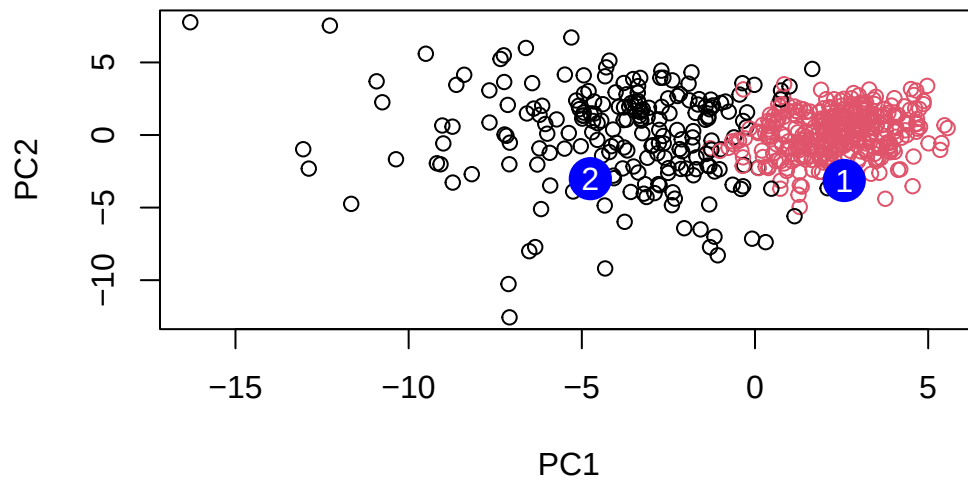
	PC15	PC16	PC17	PC18	PC19	PC20
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500

	PC21	PC22	PC23	PC24	PC25	PC26
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238

	PC27	PC28	PC29	PC30
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820

Plots the outcomes from the prediction

```
plot(wisc.pr$x[,1:2], col=grps) # plots the information of PC 1 and 2
points(npc[,1], npc[,2], col="blue", pch=16, cex=3) # assigns points
text(npc[,1], npc[,2], c(1,2), col="white") # makes labels
```



Q16. Patient 1 information should be prioritized given that they are organized amongst the cluster of malignant cases, suggesting that they may be malignant and require a followup for more detailed examination and biopsy checking. Patient 2 data lies in the benign cluster so they take lower priority for a checkup compared to patient 1.